LONDON INSTITUTE OF SPACE POLICY AND LAW

SPACE POLICY AND LAW COURSE

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SPACE HAZARDS AND SAFETY

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Planetary Protection

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Planetary Protection History

- Questions of life—the fate of life on Earth and the possibility of life elsewhere—have driven space exploration from its beginnings. Thus, planetary protection has been a concern from the start of the Space Age.

  How can we make the observations required to understand the origins, distribution, and destiny of life in the universe:

  - Without destroying or contaminating the evidence required?
  - Without changing the distribution and/or destiny of (Earth or alien) life?




PROTECTING THE EARTH
FROM THE SCUM OF THE UNIVERSE
MIB
MEN IN BLACK
COMING SOON


PROTECTING THE EARTH
FROM THE SCUM OF THE UNIVERSE
& THE UNIVERSE FROM THE SCUM OF THE EARTH
MIB
MEN IN BLACK
COMING SOON

Basic Planetary Protection Policy

- Preserve planetary conditions for future biological and organic constituent exploration
  - avoid **forward contamination**; preserve our investment in scientific exploration
- To protect Earth and its biosphere from potential extraterrestrial sources of contamination
  - avoid **backward contamination**; provide for safe solar-system exploration



The UN Space Treaty of 1967

- Article IX (part)
  - "States Parties to the Treaty shall pursue studies of outer space, including the moon and other celestial bodies, and conduct exploration of them so as to avoid their harmful contamination and also adverse changes in the environment of the Earth resulting from the introduction of extraterrestrial matter and, where necessary, shall adopt appropriate measures for this purpose."



COSPAR's Policy Categorization(s)

- Five categories of planetary protection defined by the nature of the mission to be launched and nature and understanding of the target body to be studied or encountered



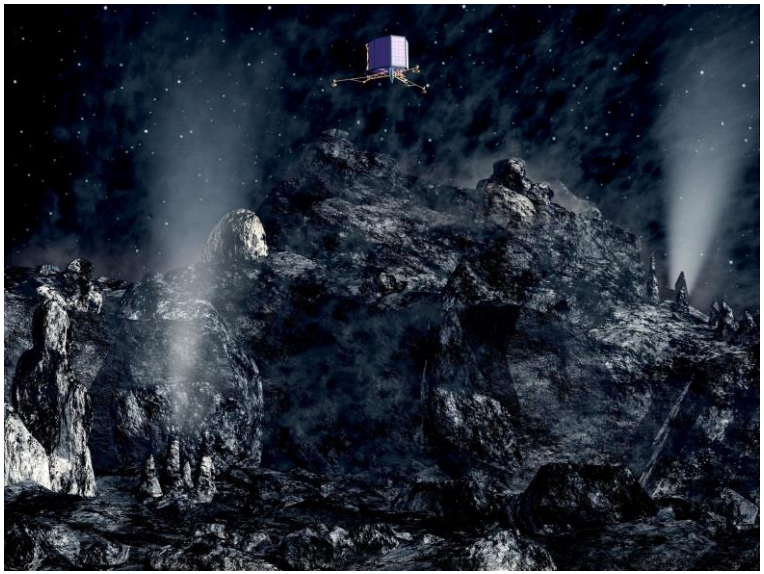
COSPAR's Policy Categorization(s)

- Category I missions—No planetary protection procedures required for missions to bodies not of interest to the study of chemical evolution and the origin of life



COSPAR's Policy Categorization(s)

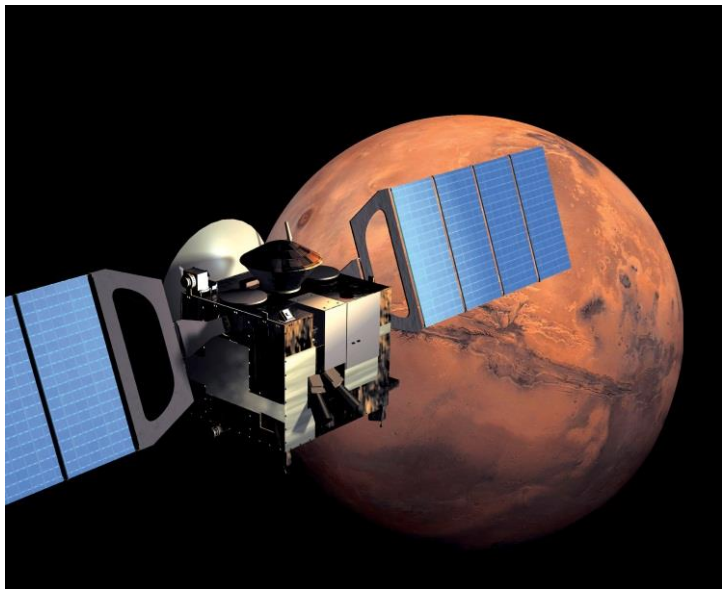
- Category II missions are those for which the target body is of interest to researchers studying organic chemical evolution and the origin of life, but where biological contamination is not thought to be possible
    - » Document spacecraft trajectories, inventory onboard organic materials, and possibly provide for the archival storage of certain spacecraft materials



COSPAR's Policy Categorization(s)

- Category III missions that fly-by or orbit planets that could be contaminated by Earth organisms
  - » Category II plus other restrictions, such as cleanroom assembly procedures and orbital lifetime restrictions







COSPAR's Policy Categorization(s)

- Category IV missions are landers to bodies that may be contaminated
  - » Category II and III requirements apply (except for orbital lifetime)
  - » Restrictions on biological contamination are generally more severe, and may include comprehensive decontamination and sterilization of the spacecraft
  - » A probability of contamination calculation is required for such missions

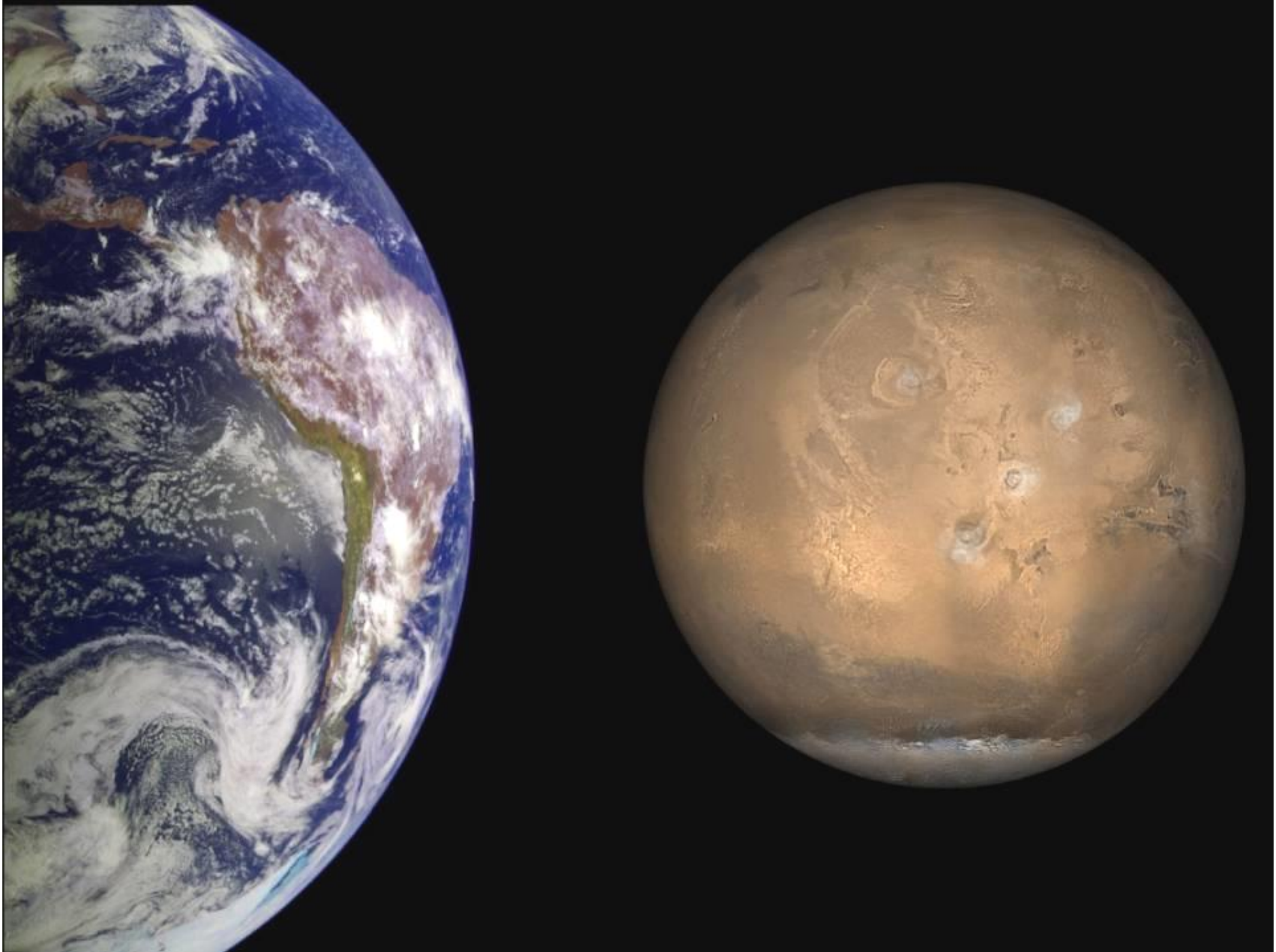


COSPAR's Policy Categorization(s)

- Earth-return missions are placed in Category V, regardless of the outbound category
  - » "Unrestricted Earth return," with no additional planetary protection requirements on the return portion of the mission
  - » "Restricted Earth return" missions, with all Category II, III, and IV restrictions plus strict controls on the handling of returned samples until their biological status is determined.

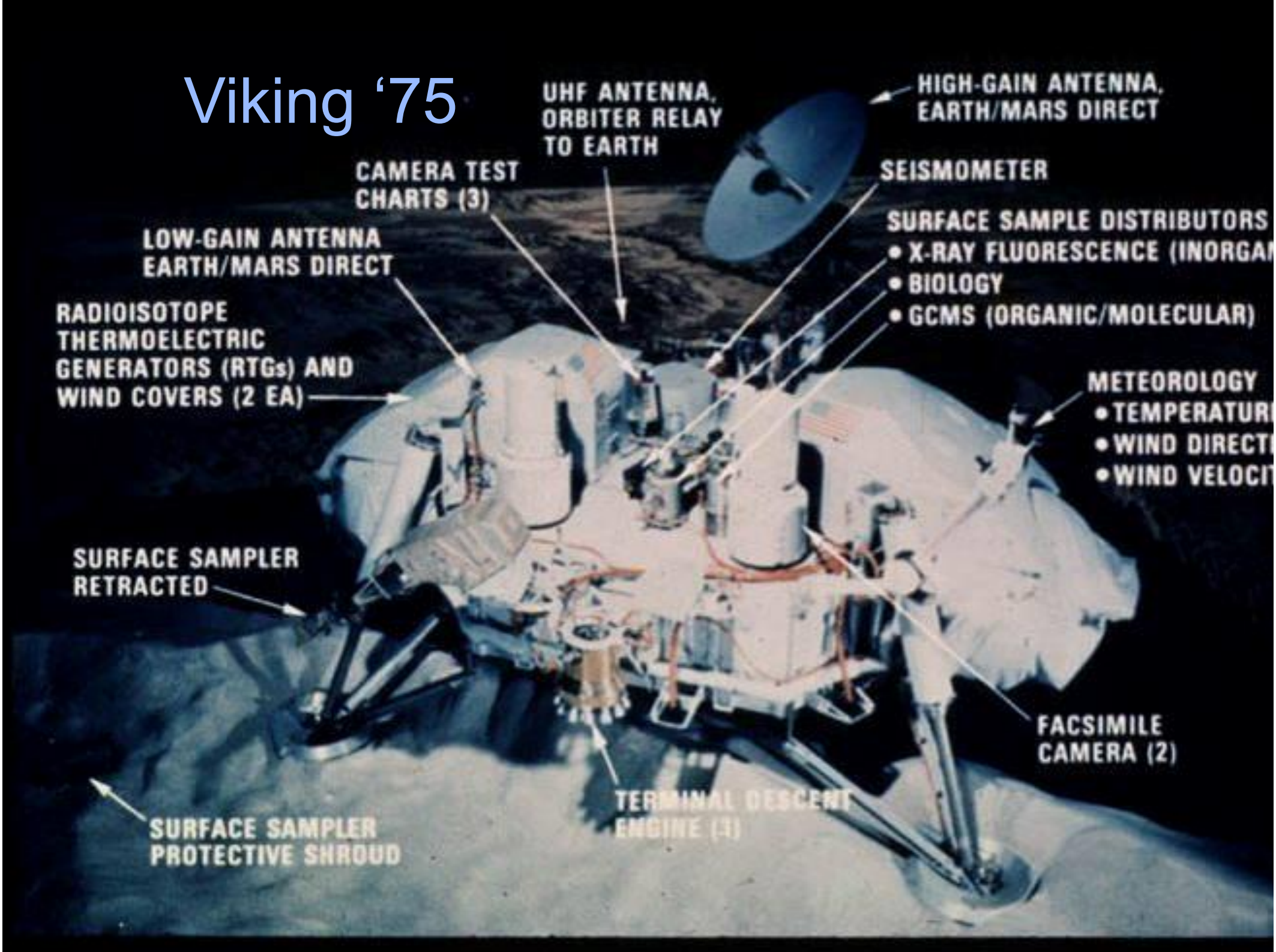








Viking ‘75


UHF ANTENNA, ORBITER RELAY TO EARTH
HIGH-GAIN ANTENNA, EARTH/MARS DIRECT
CAMERA TEST CHARTS (3)
SEISMOMETER
SURFACE SAMPLE DISTRIBUTORS
• X-RAY FLUORESCENCE (INORGA
• BIOLOGY
• GCMS (ORGANIC/MOLECULAR)
LOW-GAIN ANTENNA EARTH/MARS DIRECT
RADIOISOTOPE THERMOELECTRIC GENERATORS (RTGs) AND WIND COVERS (2 EA)
METEOROLOGY
• TEMPERATUR
• WIND DIRECT
• WIND VELOCI
SURFACE SAMPLER RETRACTED
FACSIMILE CAMERA (2)
TERMINAL DESCENT ENGINE (3)
SURFACE SAMPLER PROTECTIVE SHROUD

A Viking Casserole (DHMR)



Low humidity, each portion of the spacecraft to attain ≥ 111.7˚ C for ≥ 30 hours

Earth's Deep-Sea Hydrothermal Vents: Life-as-we-didn't know it...





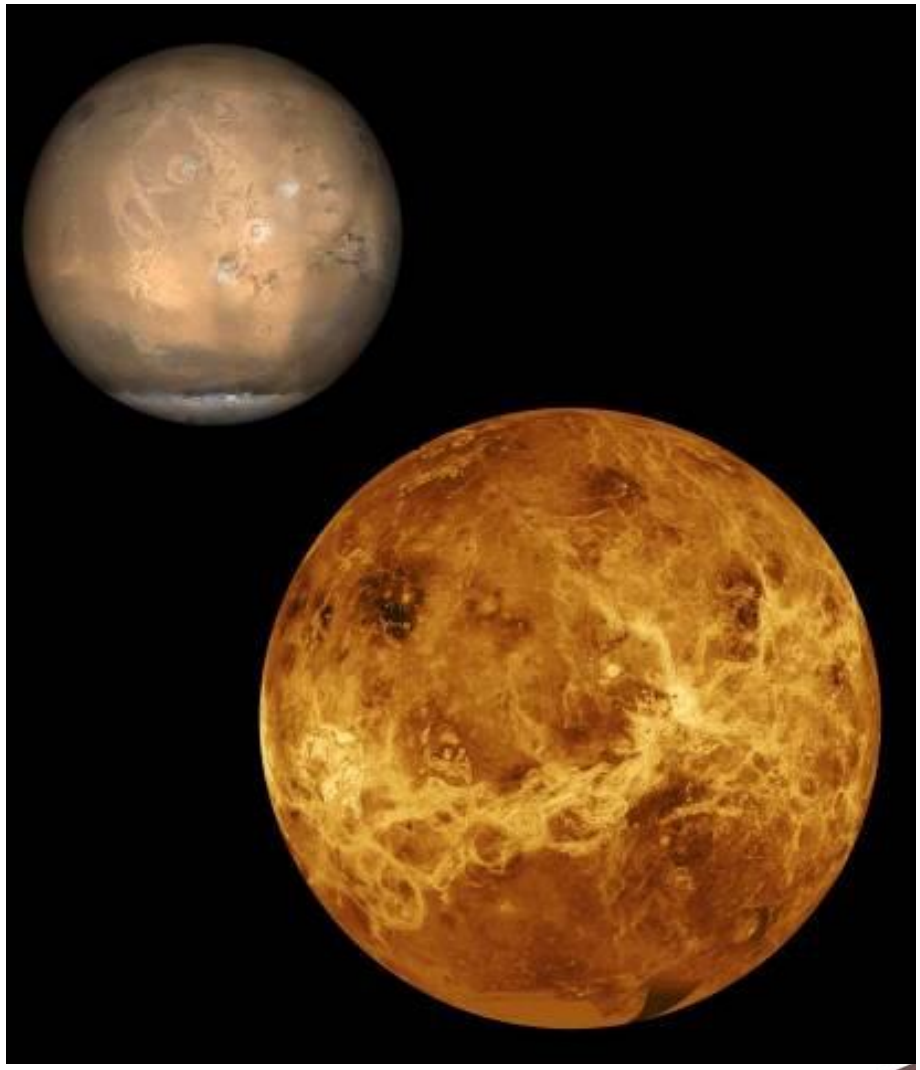


The discovery of abundant life at deep-sea hydro-thermal vents in 1977 (7 months after the Viking missions landed on Mars) surprised everybody!

- It isn't that we expect to find these things out there –
- It's that we never expected to find them *here*....

Planetary Protection: Evolving Requirements

Planetary protection evolves as the planetary exploration program progresses and as advances in planetary science—access, knowledge, measurement technologies…



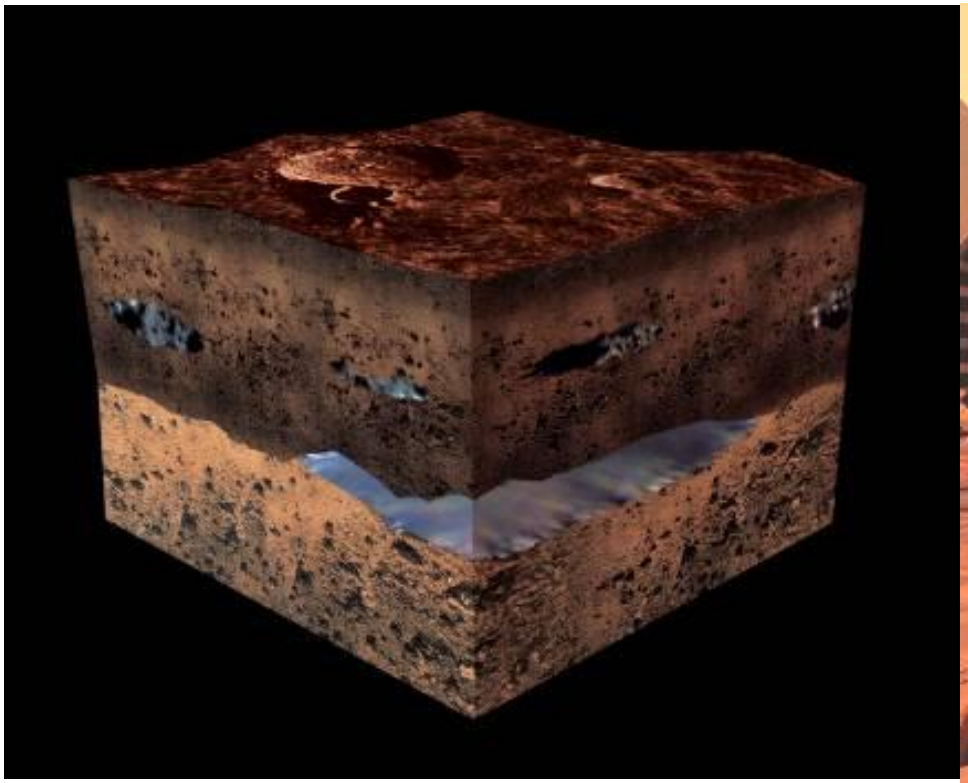




Total surface bioburden of no more than 3×10^5 aerobic bacterial spores (NSM), an average of no more than 300 spores per square meter, and total bioburden (surface, mated, and encapsulated for launch) of $\leq 5 \times 10^5$ spores

Humans on Mars?

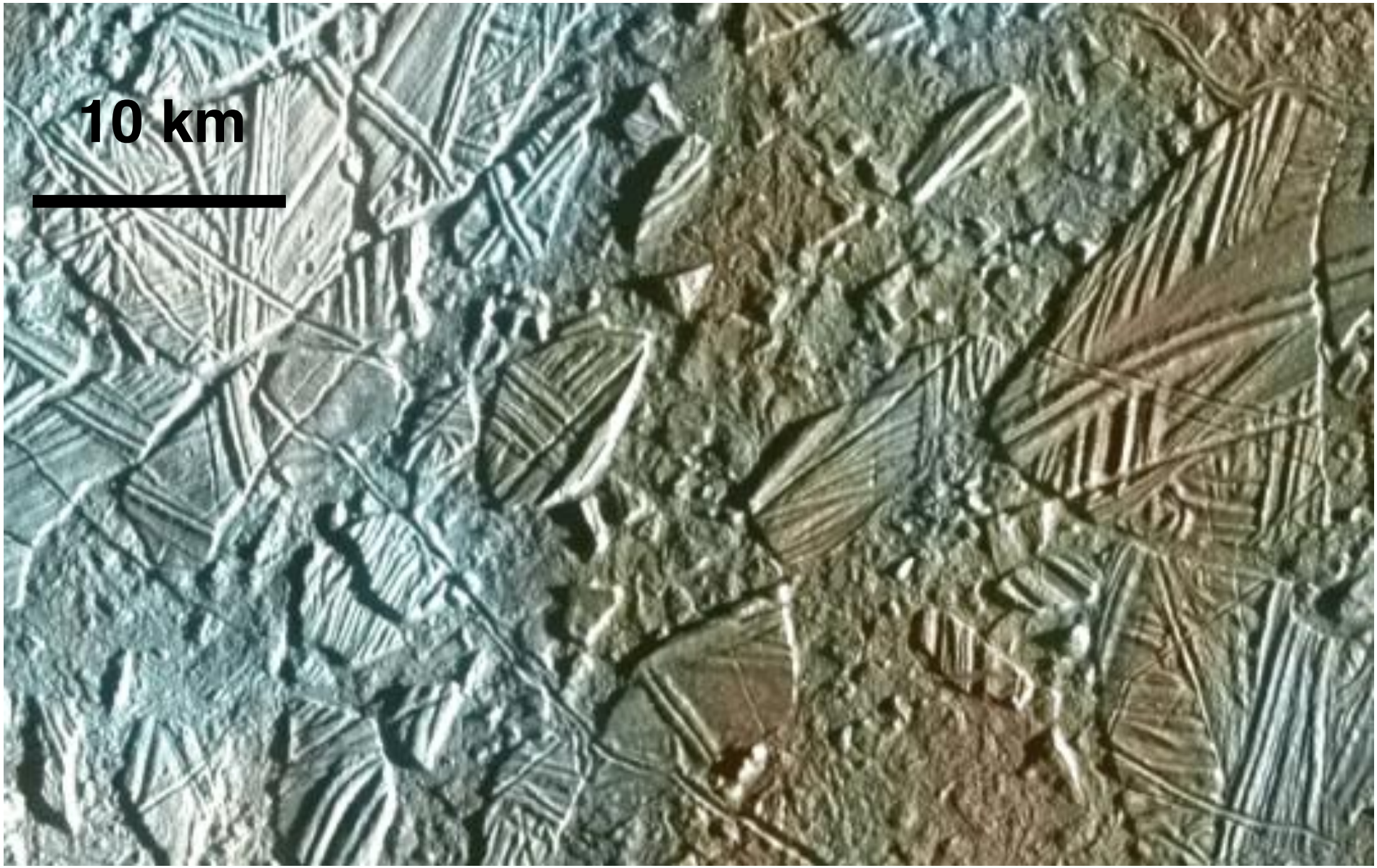
- Human capabilities may be required to explore deep subsurface aquifers, if they exist beneath the Martian deserts
- Requires a preliminary robotic search for life, because of the potential for human-associated contamination: both forward and backward





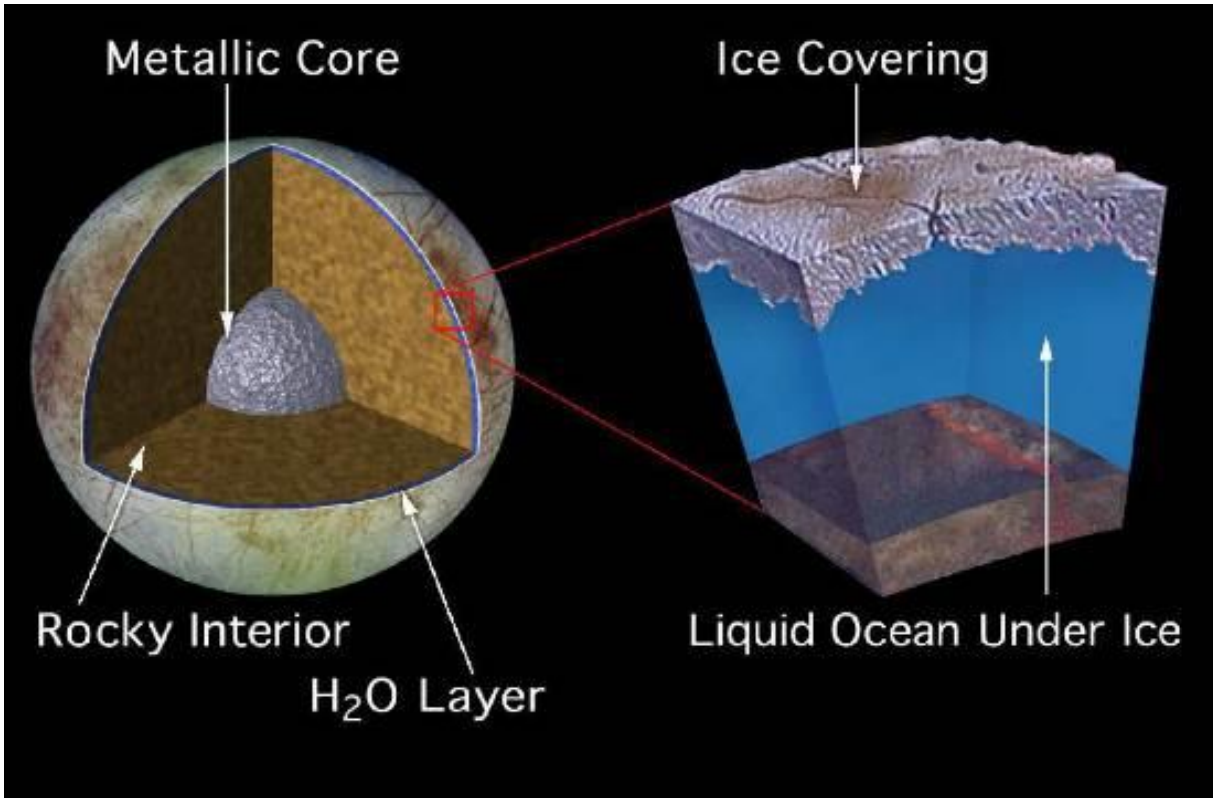


What is Europa's Ocean Like?



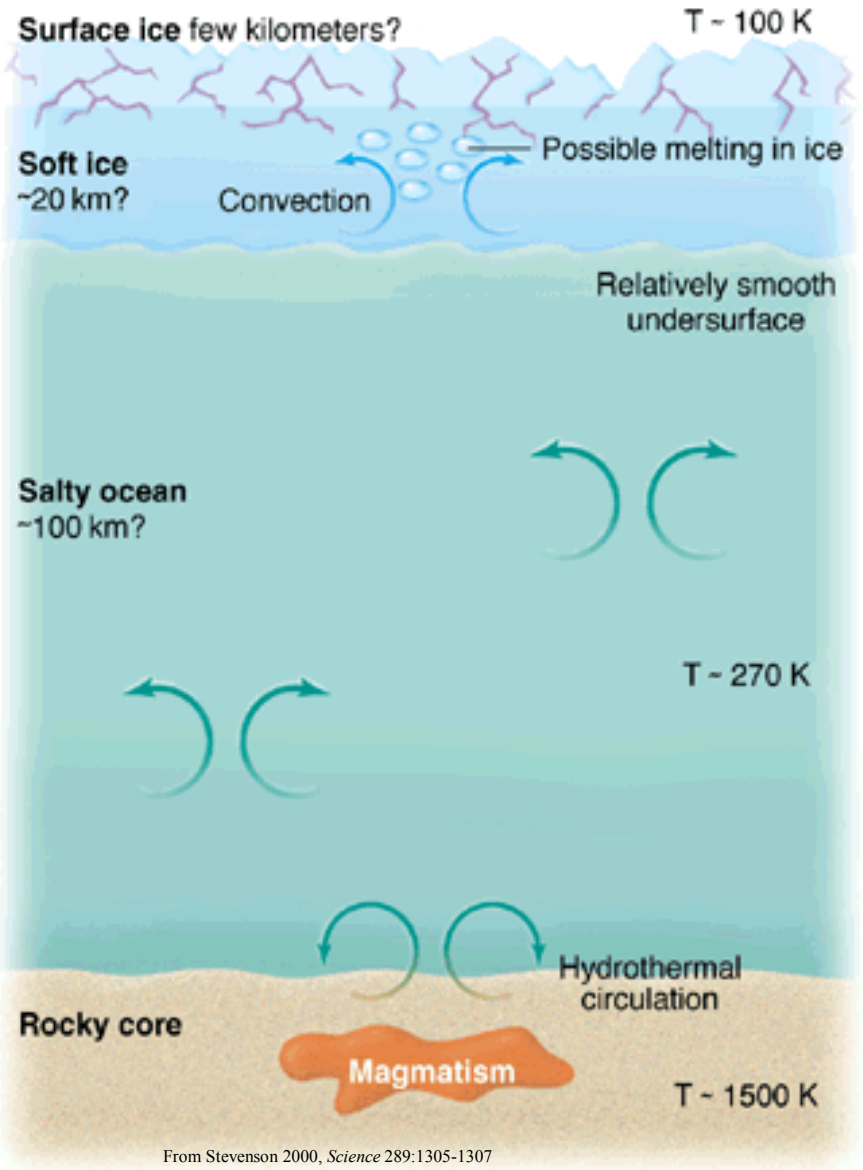
Chaos regions and "ice rafts" suggest melting and disruption of the surface

Europa—Focus of Future Astrobiological Study









From Stevenson 2000, *Science* 289:1305-1307

Hydrothermal Vents?

Current Issues in Forward Contamination Control



- Understand microbial biodiversity of cleanroom environments
- Understand survival of Earth organisms under the range of conditions that may be experienced on Mars, Europa, etc.
- Better understand the location/extent of Mars environments that may be capable of supporting Earth life, and keep track of them using a comprehensive database



- Develop and qualify new methods for the monitoring of microbial contamination on spacecraft
- Develop and qualify new methods for removing biological contamination from Mars spacecraft









UK SPACE AGENCY

Space Weather Effects

Prof Richard Crowther

Outline

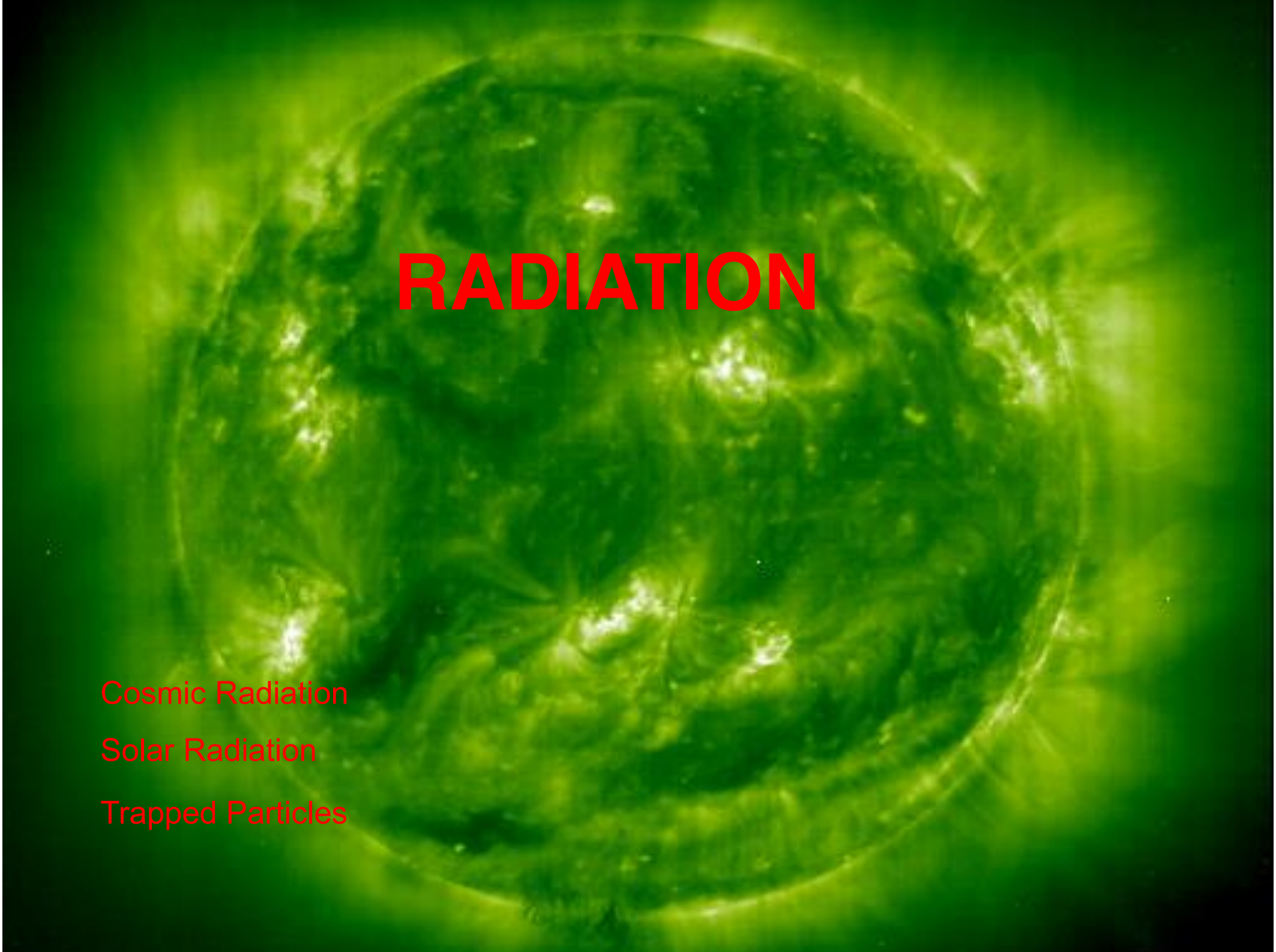
- Background
- Aviation
- Power
- Ground Transport
- Communications
- Pipelines
- Oil and Mineral Industries
- Finance



Background (Environment)

- Magnetic Storms
- Solar Radiation Storms
- Solar Radio Bursts
- Galactic Cosmic Rays
- High Speed Solar Wind
- Solar Flares



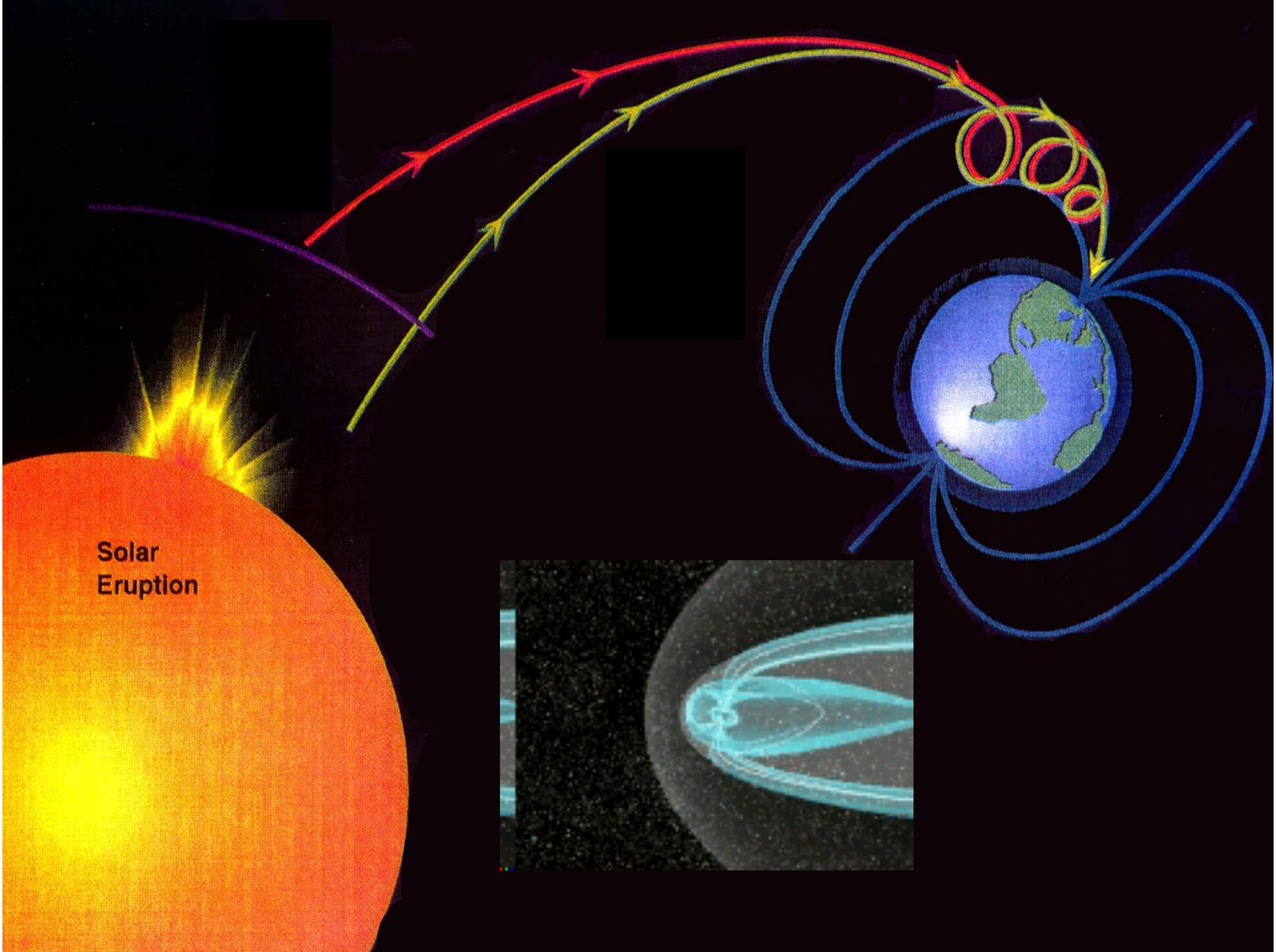


RADIATION

Cosmic Radiation

Solar Radiation

Trapped Particles


Solar
Eruption

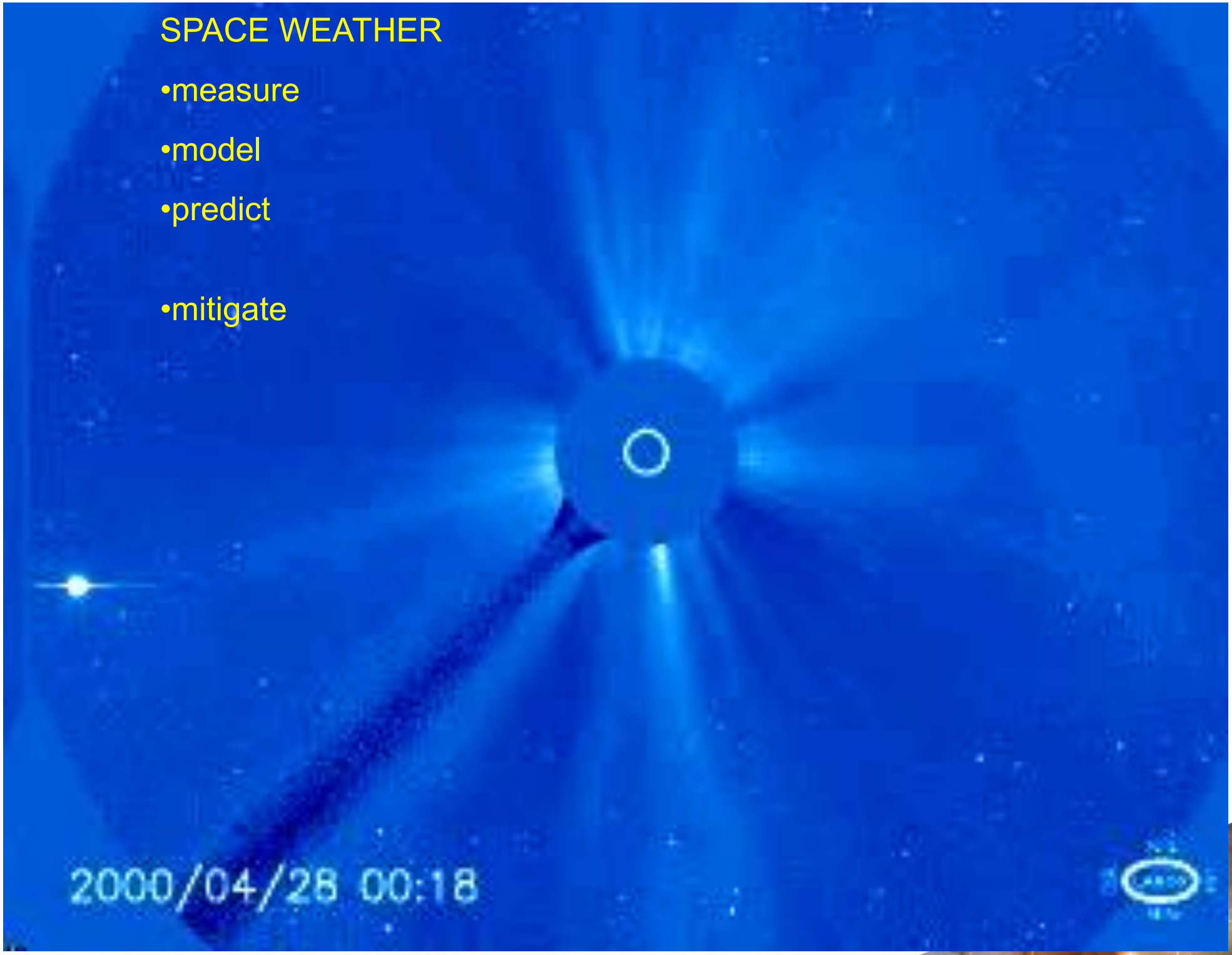


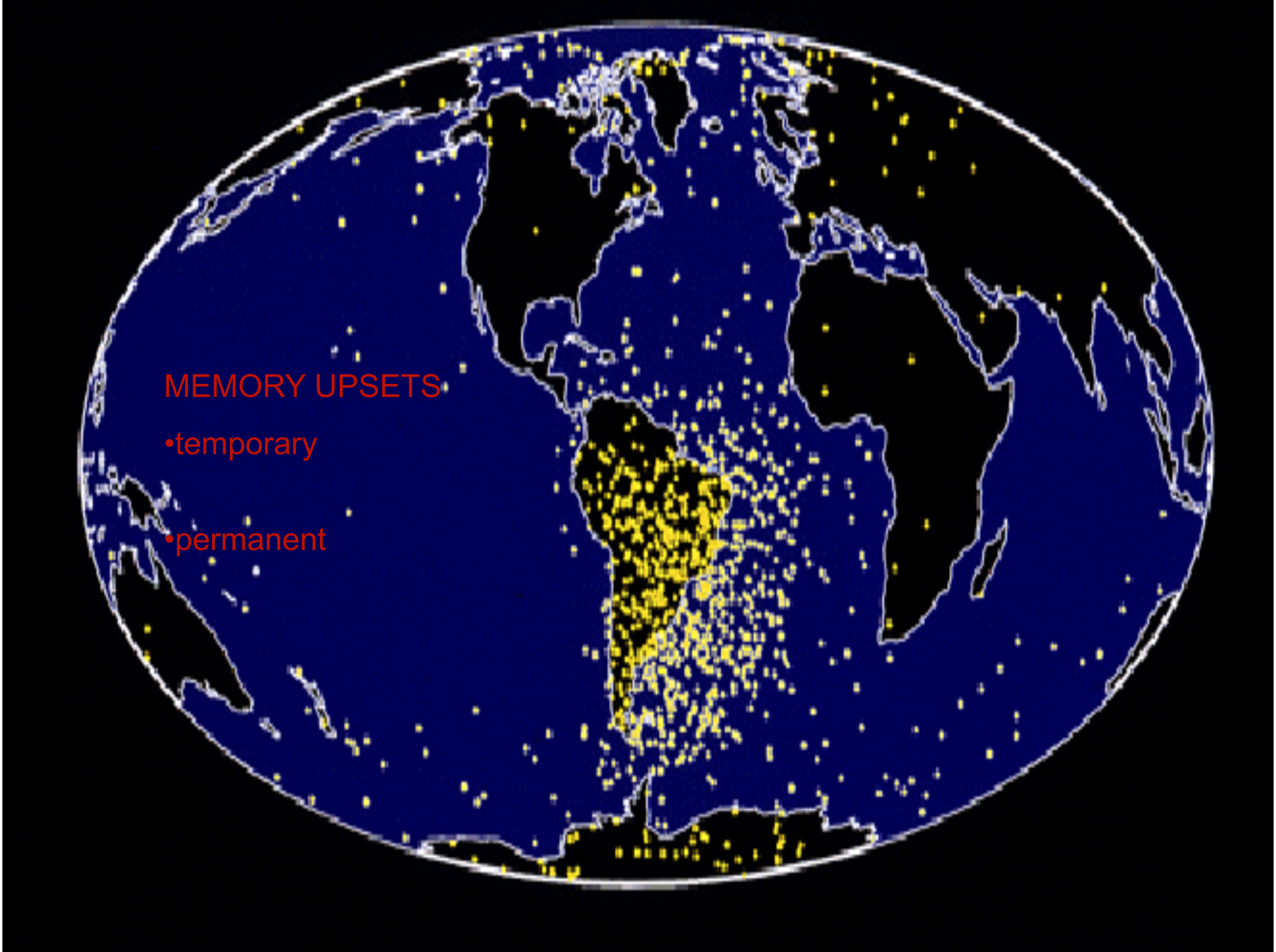


SPACE WEATHER

- measure
- model
- predict

- mitigate




MEMORY UPSETS
•temporary
•permanent



RADIATION DAMAGE

- astronauts
  - tissue damage
- spacecraft systems
  - degradation in performance
  - malfunction

Aviation

- Space weather has significant impact on Aviation, esp. trans-polar routes
- Communications
  - HF radio links can be degraded (2003, blackout)
  - GEO satcom not accessible above 82 degrees)
- Navigation
  - Accuracy can be degraded & signal lost
- Radiation Hazards
  - Compromise operation (Single Event Effects)
  - Health of aircrew and passengers





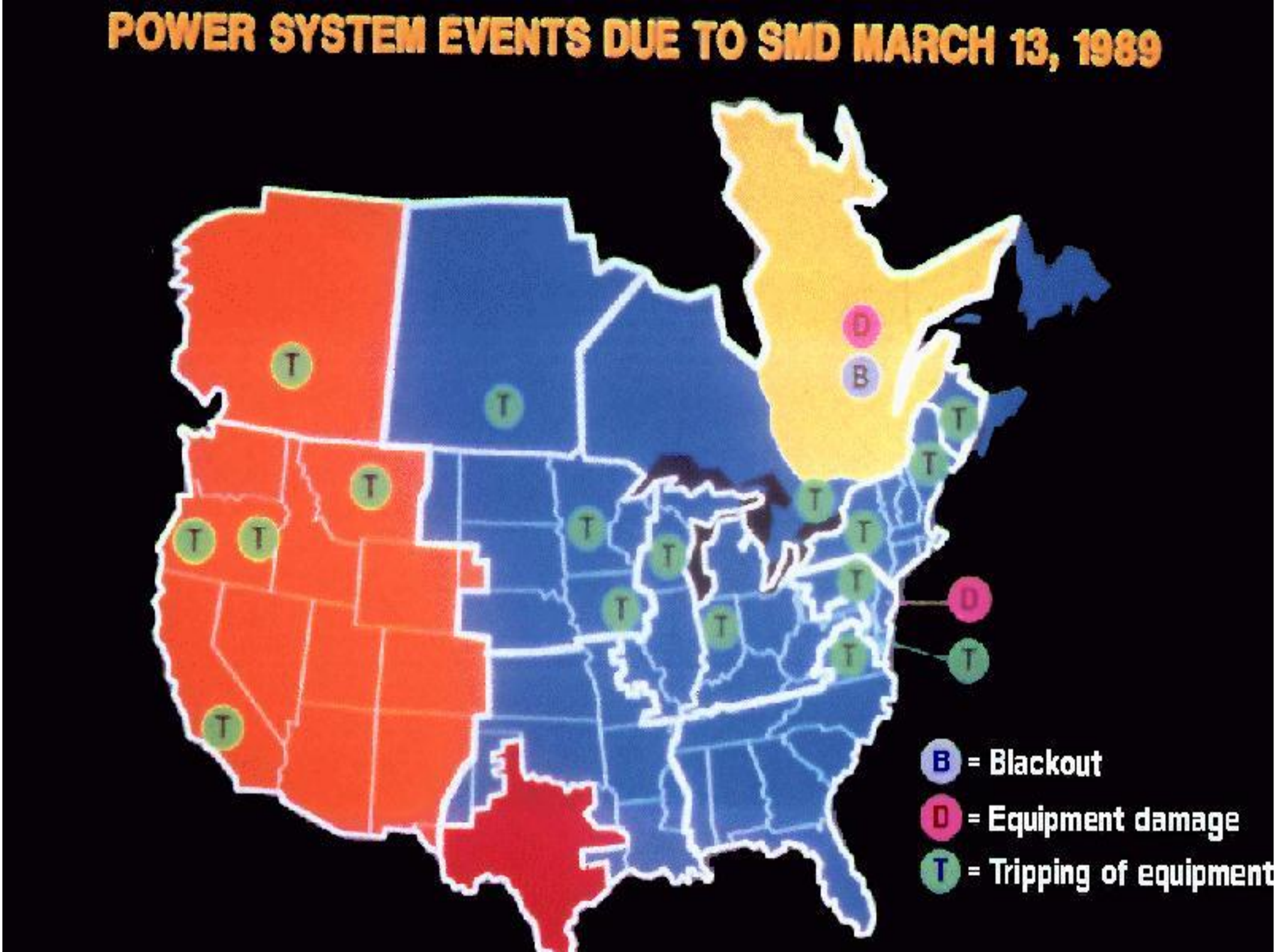

CHARGING
•Surface
•Internal

Power

- During magnetic storms, changes in Earth's magnetic field can generate sub-surface electric fields
  - Damaging currents in power grids (1989, Quebec C$2bn, New Jersey, UK)
  - Permanent damage requiring replacement
  - Cascade failure & need to restore
- Implications for "renewables" which have long transmission lines?
- Indirect consequences for fuel, food, sanitation, finance, …



POWER SYSTEM EVENTS DUE TO SMD MARCH 13, 1989





Transport

- Road and Maritime transport which use GPS less susceptible to space weather effects
  - New electronic controls may be vulnerable to SEE
- Rail traffic
  - Electric power networks (direct/indirect)
  - Currents in signalling systems
  - Control/communication technologies susceptible





Communications

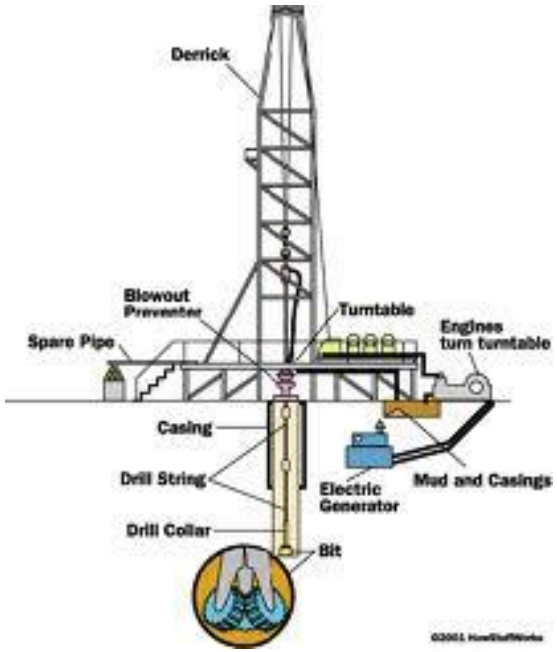


- Telephones:
  - Historic risk for:
  - Long distance systems & calls routed via comsat
  - Mobile phone links vulnerable to solar radio bursts
  - Mobile phone networks use GPS timing
- Internet is relatively robust as most traffic carried by optical fibre
- Low power wireless vulnerable to interference from solar radio bursts



Oil and Mineral Industries

- Magnetic measurements used to search for natural resources and guide drilling
- Surveys are scheduled during quiet magnetic storm conditions
- Drilling suspended during severe events to avoid misdirection ($O(10^0)$)
- Miniature magnetic sensors are reaching the consumer market





Finance

- Time stamping of financial transactions is critical to operation of markets
  - Time stamps tend to be derived from satellite navigation or internet
  - Automated trading could also be affected
  - Power and communications outages could have indirect impact



Pipelines

- Space weather can induce electrical currents (up to 1000 amps) in long metal pipelines
  - Effect is accentuated in auroral zones
  - Australian study indicates effects at moderate latitudes
- This can lead to increased corrosion rate and associated ageing & decreased lifetime





Summary

- Growth of technologies has left society more at risk from space weather
- Space weather could create major disturbances in the transport, aviation and power sectors
- GPS signals are vulnerable to space weather
- Severe space weather event could create risk to society

